GraphWalker-QA: Answering Multi-Hop Questions by Walking Entity Graphs

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Abstract

We propose GraphWalker-QA, which builds an entity graph over the corpus and answers multi-hop questions by walking from question entities to answer passages. On KaveriHop, GraphWalker-QA improves over DenseHop by 2.1 EM on 2-hop and 7.9 EM on 3-hop questions. Walking an entity graph gives larger gains on 3-hop questions than on 2-hop questions, suggesting that explicit structure helps longer reasoning chains.

Keywords: graph-based retrieval, entity graph, multi-hop question answering

1 Introduction

Dense retrievers such as DenseHop [1] reformulate queries in vector space, but each hop can drift away from the bridge entity. Building on the success of dense passage retrieval [4], we instead make bridge entities explicit: passages become nodes connected through shared entities, and retrieval becomes a walk.

2 Method

We link entity mentions in every passage and connect passages that share an entity. Given a question, we start from passages that mention the question entities and expand neighbours, scoring each step with a learned edge scorer. For KaveriHop [2] we take entity links from the dataset's entity annotations; for NarmadaQA [3] we use string matching.

3 Experiments

We compare against DenseHop [1]. On NarmadaQA we quote its published number; on KaveriHop [2] we re-run DenseHop with the released code, obtaining 48.0 and 31.9 EM, close to the numbers in the KaveriHop paper.

Method	Source	NarmadaQA 2-hop EM	KaveriHop 2-hop EM	KaveriHop 3-hop EM
DenseHop	[1]	61.3	–	–
GraphWalker-QA	ours	62.0	50.1	39.8
DenseHop	ours (re-run)	–	48.0	31.9

Table 1: Exact match on NarmadaQA and KaveriHop.

4 Analysis

The gain over DenseHop grows from 2.1 EM on 2-hop to 7.9 EM on 3-hop questions. Walking an entity graph gives larger gains on 3-hop questions than on 2-hop questions.

5 Limitations

Our walk depends on entity links. We did not measure how errors in entity linking affect accuracy.

References

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